STRhub Verified

strhub.app · strhub.app/verified

Technical Reproducibility Validation Report

gangstr

Version v2.5

Autosomal STR

Verification date	2026-11-08
Repository commit	e368b9f4dbf7ab7585036a5ff2e68480d0623c80
Result	**Does not run as documented: stops at install**
Reached	Available
Permalink	Not published — this is a test run
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Does not run as documented: stops at install**
REACHED	Available, 1/6 gates passed
WHY	The environment did not build from the declared install steps.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it does not run.

The environment did not build from the declared install steps. That does not mean the tool does not work. It means a third party following the README at this commit did not reach the end. Fix the documentation, or the step that broke, and try the link again on the new commit.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source **The run stopped here.**	FAIL
Starts	The installed program answers --help in the built environment	FAIL
Runs	Executes end-to-end without crashing *Not attempted, because the run had already stopped.*	—
Expected IO	Produces a non-empty file in the declared format *Not attempted, because the run had already stopped.*	—
Plausible Output	Output structure matches expected genotype-bearing format *Not attempted, because the run had already stopped.*	—
Execution log	—	

Stopping at a step is not a finding that the software is faulty. It records how far this particular attempt got. Software can stop early because a dependency it names is no longer available, because it expects input arranged differently from the reference sample, or because the automated environment cannot supply something it needs, such as commercially licensed components. The messages on screen when it stopped are reported below.

4. Not Built From Source

STRhub tried to build the tool from its source at the pinned commit, following the build steps the repository declares, and the build failed. Nothing below the Installs gate ran.

What this means

– **If you are trying to run it:** A build from source at this commit fails in a clean environment; the cause and a suggested fix are below.

– **If you are reviewing a paper:** Nothing ran, so this says nothing about the software's output. It records that this attempt to build it stopped, and whose side the cause is on.

– **If you maintain it:** Every cause identified sits in this run's configuration: the pinned versions, package names or build steps it declared. They are faults in how the tool was set up here rather than in the software. Re-verifying after correcting them is free.

What failed

What happened	Suggested fix
An autotools build is missing its auxiliary file: config.sub	The configure script needs 'config.sub', which `autoreconf -i` (or `automake --add-missing`) copies in from the automake package. The build runs autoconf without installing those files; running `autoreconf -fi` before `./configure` in that step fixes it.

5. Terminal Error Output

Excerpt from the execution log when the run stopped:

```
119.7 -- Installing: /opt/tool/build/thirdparty/gsl/bin/gsl-config
119.7 [ 44%] Completed 'pgsl'
119.8 [ 44%] Built target pgsl
119.8 gmake: *** [Makefile:166: all] Error 2
------
Dockerfile:14
--------------------
12 | WORKDIR /opt/tool
13 | RUN touch /tmp/.strhub_built
14 | >>> RUN cmake -S . -B build -DCMAKE_BUILD_TYPE=Release && cmake --build build -j2
15 | RUN find /opt/tool -xdev -type f -perm -u+x -newer /tmp/.strhub_built ! -path '*/CMakeFiles/*' ! -path
'*/.git/*' ! -name '*.so*' ! -name '*.a' ! -name '*.o' ! -name '*.sh' ! -name '*.py' ! -name 'config.status'
! -name 'libtool' -exec ln -sf {} /usr/local/bin/ \;
16 | ENV PATH="/opt/tool:/opt/tool/bin:$PATH"
--------------------
ERROR: failed to build: failed to solve: process "/bin/sh -c cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
&& cmake --build build -j2" did not complete successfully: exit code: 2
```

6. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · gangstr v2.5 · commit e368b9f
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( mkdir \
-p outprefix &&
GangSTR \
--bam /data/in/input.bam \
--ref /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--out outprefix \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
! \
-path '*/__pycache__/*' \
! \
-name '*.pyc' \
! \
-name '*.pyo' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

7. Reproducibility Metadata

Tool	gangstr
Version	v2.5
Repository	https://github.com/gymreklab/gangstr
Commit (pinned)	e368b9f4dbf7ab7585036a5ff2e68480d0623c80
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-11-08 14:18:43 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42488855247

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

8. Output Content Evidence

No output files were generated because the run stopped before completion.

9. Verification Data

The repository ships example data (10 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	—
Regions BED	Provided by STRhub

10. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

11. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

12. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's CMakeLists.txt (cmake), at the pinned commit. If that build fails, the published image gymreklab/str-toolkit the README points at stands in, and the report says so.
– Input: the README documents BAM (line 6); this run used STRhub's illumina-bam-hg38 reference data.
– The output directory the command writes into (outprefix) was created first.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– 1 BED path(s) replaced with /data/in/regions.bed.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.
– Regions: STRhub's ready-made gangstr file for the dataset's panel loci (STRhub records the gangstr layout for a program of this name).

13. Scope and Limitations

Concerns reproducible execution only; no claim of genotype accuracy, concordance, forensic casework suitability, or regulatory validation.

- Single reference dataset per input type
- Single containerized environment (Docker / ubuntu-22.04)
- No truth-set comparison or ground-truth genotypes
- No accuracy or concordance assessment
- No forensic validation of results
- Short-read limitations apply (very long STR alleles may not span reads)

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

Does not install

The environment for gangstr v2.5 could not be built from the declared install steps, so nothing further was attempted.

No output produced

No file in the declared format was written, so nothing could be checked for structure or content. This says the run did not get that far; it is not a finding about the quality of the software.

Verdict: Fails

The environment did not build from the declared install steps.

Repository test data not used

The repository ships example data (10 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

Reproducibility statement

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.